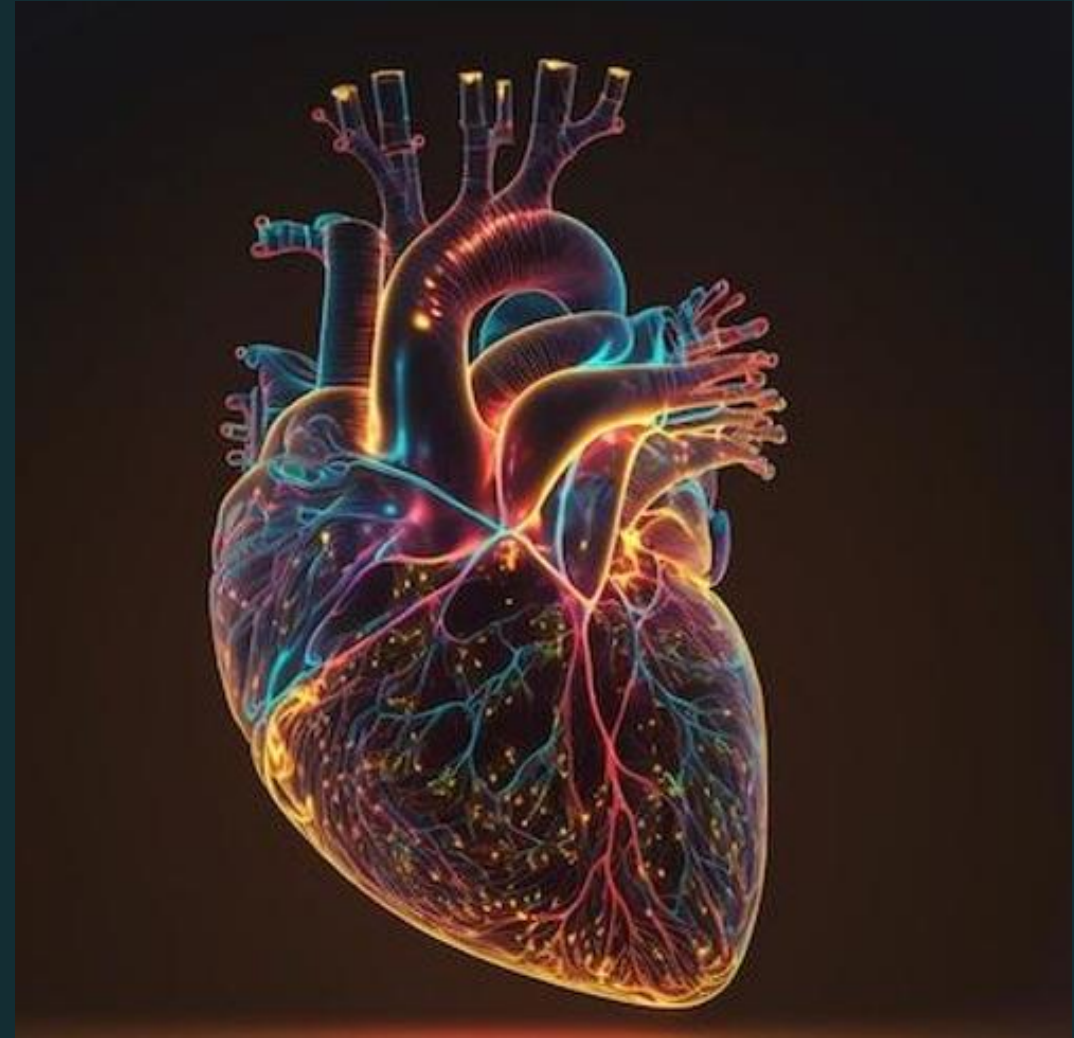


GEOGRAPHIC ACCESS TO CARDIAC CARE

# Outcomes Across the Southern U.S.

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## WHY THIS PROJECT

Several years ago, I was on a golf course with my husband and his friends when I noticed one of them rubbing his arm and grimacing, signs I'd seen hundreds of times in the echo lab.

We were in a small town. Our cell service kept dropping. The nearest hospital, 45 minutes away, we found out it wasn't equipped for cardiac care after spending an hour or more there, so we had to call 911 again from the hospital and wait another 45 minutes for transfer to the second hospital.

The cardiologist later told us: a few more minutes, and he might not have survived.

*My story is about emergency access: can someone get care fast enough, in the moment, in a rural community? I found out that EMS response-time data isn't tracked nationally at the county level. So I went looking for the closest thing that is measured everywhere: readmissions, after someone is diagnosed.*



# How does geographic access to cardiac care influence readmission outcomes?

*Note: readmission tracks care after diagnosis, not how fast someone first reached a hospital (an EMS question, outside this dataset).*

**01**

## Shortage vs. outcomes

Does cardiac care shortage status correlate with readmission rates?

→ Slide 6

**02**

## Social determinants

Do income, insurance coverage, and rurality predict readmission performance?

→ Slide 7

**03**

## Hidden winners

Which high-shortage counties outperform expectations, and why?

→ Slide 8

## METHODOLOGY

**13 states. 1,318 counties. Four public datasets.**

1,318

Southern U.S. counties analyzed across AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, TX, VA, WV

- **HRSA HPSA**  
Cardiac & primary care shortage designations
- **HRSA MUA/P**  
Medically underserved areas & populations
- **CMS Readmissions**  
30-day hospital-level cardiac readmission rates
- **County Health Rankings**  
Income, insurance coverage, rurality

## TOOLS

### Python

Cleaning, merging, EDA, correlation testing

### Power BI

Interactive county-level dashboard

## Can we even see cardiac care access in most counties?

*What I found is that most of the rural counties lack proper cardiac data to asses.*

# 86.1%

**Almost 9 in 10 counties with no cardiac care are rural.**

I checked: 837 of 1,318 Southern counties have no cardiac readmission data on file, not because they had zero readmissions, but because no hospital there met the minimum case count CMS requires to report. When I looked at who those 837 counties were, 721 of them turned out to be majority-rural. So the counties most likely to be cut off from care are also the ones we have no way to measure.



## Does a shortage designation predict how a county's patients fare?

*No. Two different shortage scores were tested, and neither reliably predicted cardiac readmission outcomes.*

### HPSA Score

*measures primary-care shortage severity*

0.079

**Barely any connection**

I compared shortage severity to readmission outcomes across every county. If shortage mattered a lot, worse-shortage counties would show worse outcomes. They mostly didn't.

*(Could easily be due to chance)*

### IMU Score

*Index of Medical Underservice - measures how underserved an area is*

0.047

**Almost no connection**

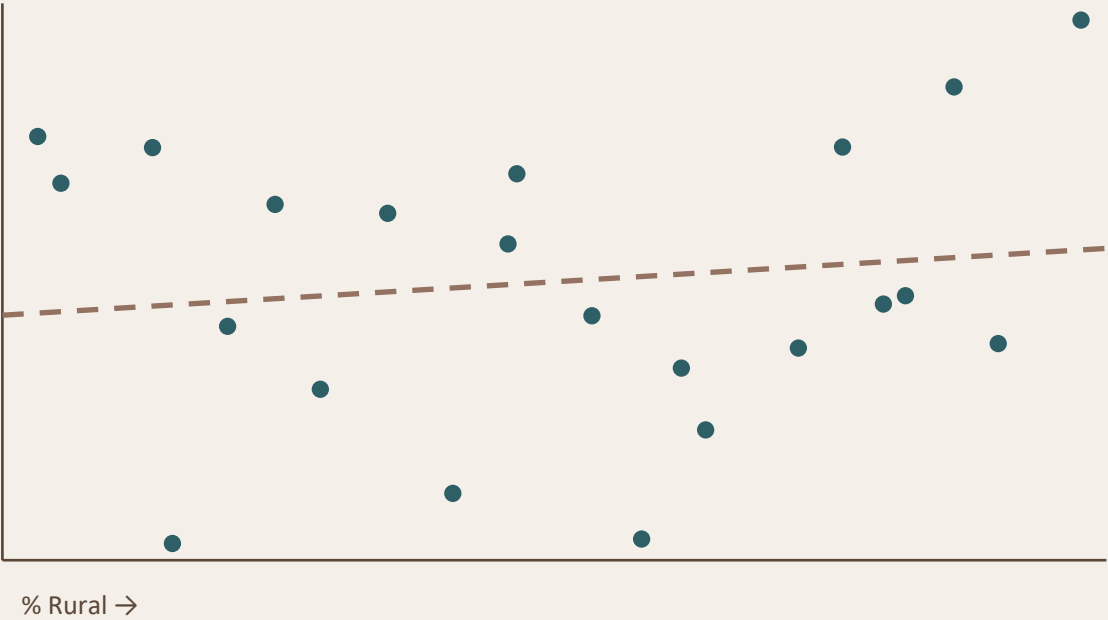
Same test, using a different underserved-area measure instead of shortage score and severity of the designation didn't track with outcomes.

*(Could easily be due to chance)*

# Does a community's income or rurality predict its access to effective cardiac care?

*Income and rurality showed almost no link to readmission outcomes, the same result we saw with shortage scores in Question 01.*

Rurality vs. Readmission Performance (pattern matches  $r = 0.090$ )



0.090

Almost No Connection

I compared rurality levels across every county to their cardiac readmission rates. If rurality mattered a lot, more rural counties would show consistently worse outcomes. They didn't.

*(Statistically real, but too small to act on alone)*

## Do all high-shortage counties struggle with cardiac outcomes?

*33 counties beat the odds, ranking among the best despite a severe shortage.*

# 33

High-shortage counties that still perform in the top tier for readmissions.

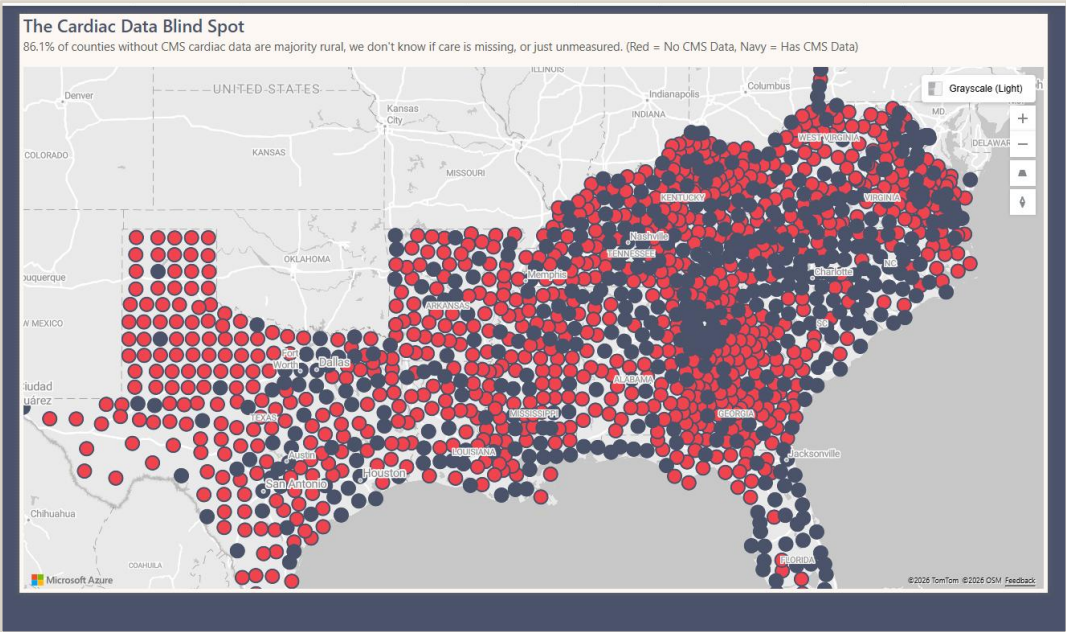
**Not just rural:** % Rural ranges from under 1% (Cobb County, GA) to 100% (Letcher County, KY), roughly split between urban and rural.

### Why it matters

A shortage score can come from just one underserved facility, not the whole county. That's why severity and outcomes don't always line up. These 33 counties are outperforming their risk profile anyway.

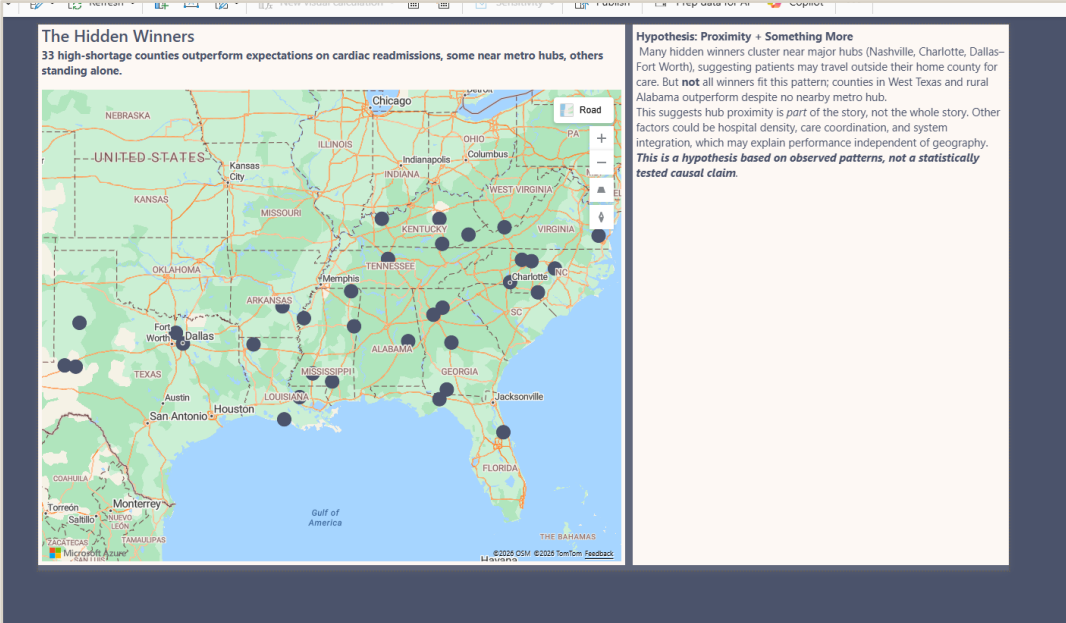


# Interactive views — explored live



The Cardiac Data Blind Spot

Red = no CMS data · Navy = has CMS data



The Hidden Winners

33 high-shortage counties outperforming expectations

## TAKEAWAYS

# Fix the map before fixing the outcome.

1

### Close the visibility gap first

86.1% of "missing data" counties are rural. Extend reporting thresholds or supplemental surveillance before drawing conclusions from CMS data alone.

2

### Don't rely on shortage score as a proxy

Neither HPSA nor IMU scores predict readmission outcomes. Don't use shortage score alone to decide where resources go.

3

### Study the Hidden Winners

33 counties are beating their risk profile. Their care models could be a playbook for other counties.



# Thank You

Geographic Access to Cardiac Care: Outcomes Across the Southern U.S.

